CLIENT REVIEW DOCUMENT

Rahma AI Guardian App Architecture and Implementation Plan

A detailed Phase 1 and Phase 2 plan covering system architecture, emergency flows, offline fallback, server architecture, open technical options, recommended decisions, and launch readiness.

PREPARED: 22 MAY 2026

Recommended path

Native iOS and Android apps, AWS UAE Region with Bedrock if approved, constrained on-device fallback, SMS fallback, reusable backend services, and a receiver dashboard.

EXECUTIVE RECOMMENDATION

Build Phase 1 as a controlled emergency-care pilot first.

Rahma should first prove one reliable path: activate, locate, identify basic context, alert caregiver or receiver, support hospital/emergency handoff, log every action, and continue safely when connectivity is weak.

Native Mobile

Better control over permissions, background location, SMS fallback, haptics, voice, and future wearables.

Client-Approved Hosting

Production data, logs, backups, and keys stay inside the approved hosting boundary unless formally approved otherwise.

Evidence-Ready Controls

We will meet any compliance, audit, data handling, or security requirement specifically confirmed by the client.

CONFIRMED SCOPE

Phase 1 focuses on the emergency foundation.

One-Touch Emergency Activation

Large press-and-hold button with vibration, sound, visual confirmation, false-alarm handling, and safe escalation.

Location and Profile Packet

Current GPS, network location, last known location, timestamp, confidence, language, medical flags, allergies, and caregiver contacts.

Caregiver and Receiver Alerts

Push and SMS fallback with secure short-lived links, acknowledgement tracking, retry rules, and receiver dashboard status.

Hospital Finder

Approved Dubai and Abu Dhabi facility directory with nearest emergency-ready facility suggestions and explainable routing.

Medication Reminders

Timed reminder notifications, simple taken/skipped flow, and optional caregiver missed-dose escalation.

Arabic and English Flow

Bilingual prompts, simple voice intent support, no-response fallback, and reviewed emergency wording.

HOW THE APP WORKS

From button press to emergency handoff.

The app should be usable by an elderly or special-care user without typing or menu navigation. A 2-second press-and-hold starts the emergency path, while voice, visual buttons, location fallback, and SMS fallback help the system continue safely when conditions are imperfect.

How Rahma Works in Phase 1

1. User presses and holds the Rahma button for 2 seconds	2. App asks one simple voice/visual prompt
3. App captures current or last-known location	4. Minimal emergency profile is attached
5. Backend creates a timestamped emergency event	6. Caregivers and receiver dashboard are alerted
7. Approved authority API, dashboard, or manual handoff starts	8. Receiver actions and delivery status are audited

Authority dispatch recommendation

For Phase 1, do not make launch dependent on a live authority API unless that API is already approved and testable. Our recommended method is to build the emergency event packet and receiver dashboard first, support approved manual/call-center handoff or a simulator during pilot validation, and switch to direct authority dispatch once the client confirms the integration route.

USER ONBOARDING

Onboard safely without making elderly users do all the work.

Our recommendation is to use UAE Pass as the primary verified identity route where approved, assisted onboarding as the elderly friendly fallback, OTP plus relationship approval for caregivers, and client SSO for staff/dashboard users.

Recommended Phase 1 onboarding flow

1. Pilot invite or staff-assisted registration starts the process.
2. User chooses onboarding path: UAE Pass, assisted onboarding, or approved manual pilot setup.
3. Identity is verified and a Rahma user reference is created.
4. Minimal emergency profile is completed: language, medical flags, allergies, caregiver contacts, and medication reminders.
5. Caregivers receive invite and verify by OTP plus relationship approval.
6. App performs permission health check for location, notifications, microphone, and SMS fallback.
7. User completes a guided emergency test in training mode.
8. Account becomes pilot-ready only after profile, caregiver, permissions, and test status are complete.

UAE PASS

Best fit: Best for verified patient/user identity when the client confirms access and user suitability.

Condition: Requires integration access, user readiness, UAE Pass app availability, and a fallback for users who cannot complete it.

Our recommendation: Recommended primary route for Phase 1 users.

ASSISTED ONBOARDING

Best fit: Best for elderly or special-care users who need help from pilot staff, clinic staff, or caregivers.

Condition: Requires operational support and staff verification steps, but reduces user drop-off.

Our recommendation: Recommended fallback and pilot-support route.

MOBILE OTP

Best fit: Best for caregivers and family members who only need verified phone access and relationship approval.

Condition: Not strong enough alone for sensitive patient identity; should be paired with relationship verification.

Our recommendation: Recommended for caregivers, not as the only patient identity method.

CLIENT / STAFF SSO

Best fit: Best for admins, support teams, receiver dashboard users, and authority/partner staff.

Condition: Depends on client identity provider readiness and role mapping.

Our recommendation: Recommended for all staff and dashboard access.

Recommended Phase 1 Onboarding Flow

Pilot invite or assisted registration

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Choose onboarding path: UAE Pass, assisted, caregiver OTP, or staff SSO

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Create Rahma user reference or approve dashboard access

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Complete emergency profile and medication reminder details

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Invite and verify caregivers

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Run permission health check

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Complete training-mode emergency test

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Pilot-ready account

RECOMMENDED APP STACK

Use true native apps for the safety-critical mobile experience.

Rahma depends on background location, emergency fallback, haptics, push delivery, encrypted local data, accessibility, voice prompts, and future wearables. Those are exactly the areas where native iOS and Android implementation gives us the most control.

IOS APP

Swift + SwiftUI

Best fit for a safety-critical iPhone app because it gives direct access to Core Location, PushKit/APNs, haptics, Keychain, background modes, HealthKit/watchOS expansion, and Apple accessibility APIs.

ANDROID APP

Kotlin + Jetpack Compose

Best fit for Android because it works closely with Android permissions, foreground services, WorkManager, background location, SMS intent behavior, encrypted storage, wearable APIs, and Material accessibility patterns.

SHARED APP STANDARDS

Common API contract + shared design rules

Keep iOS and Android consistent by using the same backend request/response definitions, emergency status names, event fields, colors, typography, spacing, and accessibility rules.

UX SYSTEM

Native accessibility-first design system

Large controls, high contrast, Arabic/English typography, haptic feedback, voice and tap in parallel, emergency status banners, and no typing in critical flows.

OFFLINE AND SMS STACK

Design offline mode as a safe degraded emergency path.

Offline mode should not promise the full Rahma product without internet. It should guarantee that emergency activation, local profile lookup, last-known location, SMS fallback, and event sync are handled safely.

OFFLINE VOICE

Use platform STT first; test WhisperKit / whisper.cpp as fallback proof

For Phase 1, offline voice should only handle simple commands, language hints, and yes/no context. It should not decide whether an emergency is real.

OFFLINE DATA

Encrypted local cache + event queue

Store only minimal emergency profile, caregiver contacts, last known location, and unsynced event logs using Keychain/SQLCipher or encrypted platform storage.

SMS FALLBACK

Primary: AWS End User Messaging/SNS; backup: Twilio or Infobip

Use registered sender IDs, pre-approved emergency templates, delivery callbacks, retry rules, and a backup provider for high-availability fallback.

ON-DEVICE RULES

Rules engine before AI

Button activation, fall/silence, high-risk profile flags, and missing connectivity should trigger safe escalation even if speech recognition fails.

Recommended SMS provider order

Primary: AWS End User Messaging SMS / SNS if the backend is hosted on AWS UAE, because it keeps provider operations close to the cloud stack and simplifies monitoring.

Secondary: Twilio or Infobip as a backup route if the client wants multi-provider failover, better regional routing coverage, or a separate messaging vendor.

Condition: UAE SMS should use registered sender IDs and approved templates. Emergency SMS content should be short, pre-approved, and avoid unnecessary sensitive data.

Fallback message model

SMS should include the user reference, emergency status, timestamp, last/current location text, and a secure short-lived link only if URLs are approved for the sender registration.

If URLs are not approved, the SMS should send a plain location coordinate or last known area and ask the caregiver/receiver to open the dashboard.

HIGH-LEVEL ARCHITECTURE

Mobile-first platform with reusable Phase 2 foundations.

The architecture separates mobile experience, receiver interfaces, backend services, AI/voice support, data storage, and partner integrations so Phase 2 can expand without rebuilding the core.

High-Level Architecture

Native mobile app one-touch, voice, offline fallback	Receiver interfaces caregiver, support, admin
API gateway auth, validation, routing	Core backend services identity, consent, emergency events
Operational services location, notifications, reminders, hospitals	Data layer encrypted DB, queues, audit, monitoring
Integration adapters SMS, push, maps, emergency handoff	Phase 2 partners schools, pharmacies, wearables

EMERGENCY AND OFFLINE FLOWS

Help must start even when the internet is imperfect.

The offline path should not repeat the normal emergency flow. It should show exactly how Rahma behaves when internet, GPS, push notifications, or authority APIs are unavailable.

Offline and Weak-Network Flow

Activation succeeds: 2-second hold, silent panic, or future wearable trigger

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Create local emergency record and load encrypted profile

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Attach fresh GPS/network location, or last-known location with low-confidence timestamp

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If internet is available: send event packet to backend and alert caregiver/dashboard

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If approved authority route is live: send emergency packet through approved route

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If authority route is pending: use receiver dashboard, manual handoff, or simulator

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If internet is unavailable: send SMS-safe payload to approved caregivers or approved handoff number

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Queue encrypted event locally and sync timeline, delivery results, and diagnostics when connection returns

OPEN OPTIONS

Recommended decisions for unconfirmed items.

TOPIC	OPTIONS	RECOMMENDED BY US
Mobile technology	Native iOS/Android, Flutter, React Native	Native iOS and Android for stronger device, location, SMS, background, and wearable control.
Hosting	AWS UAE Region + Bedrock, custom on-prem, hybrid	AWS UAE Region + Bedrock unless the client mandates dedicated hosting. Exact model and feature availability should be confirmed during kickoff.
AI model usage	Managed private AI, self-hosted model, public AI API	Managed private AI in approved hosting. Public AI only for non-sensitive prototypes

		if formally approved.
Offline voice	Lightweight on-device model, rules-only prompts, full offline AI triage	Lightweight on-device model for simple commands and yes/no; no full offline medical triage in Phase 1.
Emergency integration	Direct approved API, gateway/call-center, simulator/manual pilot	Use approved API if available; keep simulator/manual handoff ready so pilot is not blocked.
Onboarding	UAE Pass, mobile OTP, client SSO, assisted onboarding	UAE Pass if client confirms access; OTP or assisted verification as fallback.

IMPLEMENTATION PLAN

16-week Phase 1 delivery timeline.

Weeks 1-2 — Discovery and Scope Lock

Final scope, hosting decision, onboarding decision, emergency handoff path, UX wireframes, data inventory, and risk register.

Weeks 3-6 — Core App and Backend MVP

Native app MVP, profile service, emergency service, caregiver flow, notifications, hospital directory, and dashboard MVP.

Weeks 7-9 — Offline and Emergency Reliability

SMS fallback, local event queue, last-known location, receiver lifecycle, false-alarm handling, and poor-connectivity tests.

Weeks 10-12 — AI, Security, and Script Validation

Arabic/English prompts, confidence policy, reviewed scripts, access testing, audit review, and support runbook.

Weeks 13-16

Pilot Readiness and Launch

Production deployment, pilot onboarding, dashboards, incident process, support training, and Phase 2 backlog.

PHASE 2 READINESS

Design Phase 1 so children, geofencing, pharmacy, and languages fit later.

Flexible Roles

Add child, parent, school, pharmacy, and support roles without changing the identity model.

Location Service

Emergency-only location in Phase 1, safe zones and geofencing in Phase 2.

School Workflows

Parent/school permissions, escalation rules, and authorized receiver dashboards.

Localization Pipeline

Versioned prompts and reviewer workflow for additional languages.

SERVER ARCHITECTURE

Preferred AWS UAE Region architecture if approved.

Preferred Server Architecture

iOS / Android apps and dashboards

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WAF + API Gateway

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Private VPC

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Application services: containers + serverless workers

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Encrypted data services: relational DB, cache, queues, object storage

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Managed AI layer: Bedrock if approved and available

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Key management, secrets, backups, monitoring, logs, and audit events

Custom On-Prem

Maximum control, but higher cost, procurement, GPU sizing, MLOps, patching, and 24/7 operations.

Hybrid

Managed app services plus private AI server. Useful if AI workloads need stricter isolation.

Offline Boundary

Offline mode covers emergency-critical fallback, not the full product experience.

RISK HANDLING

Expected risks and mitigation.

Emergency API access is delayed

Build simulator and manual handoff paths in parallel.

AI misunderstands distressed speech

Button activation starts the emergency flow regardless of AI confidence.

Weak GPS indoors

Use GPS, network location, last known location, timestamp, and confidence.

Internet unavailable

Use local event creation, SMS fallback, receiver dashboard/manual handoff, and later sync.

Caregiver access misconfigured

Verify caregivers, use RBAC, short-lived links, and audit access.

Custom hosting required

Prepare a separate infrastructure proposal if the client rejects managed cloud.

CLIENT DECISIONS NEEDED

Items to confirm before Phase 1 build lock.

- Hosting preference: AWS UAE Region with Bedrock, dedicated on-premises infrastructure, or hybrid.
- Onboarding preference: UAE Pass, mobile OTP, client SSO, or assisted/manual onboarding.
- Emergency handoff model for Phase 1.
- SMS payload approval for fallback alerts.
- Receiver roles for Phase 1 alerts.
- Approved hospital directory source.
- Medication reminder scope for Phase 1.
- Voice/audio retention policy.
- Phase 2 priority order.
- Any specific audit, security, data residency, or documentation requirements to be met.

Final recommendation

Proceed with a disciplined Phase 1 pilot focused on emergency reliability, accessibility, location capture, caregiver/receiver alerting, medication reminders, auditability, and operational readiness. Approve Phase 2 after Phase 1 validates real-world activation, alert delivery, location quality, false alarm management, support capacity, and the agreed hosting/requirement model.

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